Install Manual
Pi Zero 2 W

How to install, update and look after the Node Controller web tools on a 512 MB Raspberry Pi Zero 2 W running AllStarLink with DVSwitch.

DASHBOARD · SYSMON · INSTMON · WIFIMON · 44HELPER · NODE TALKER · NODE RECORDER · PIBRARY · TERMINAL MONITOR

FILES COVERED BY THIS EDITION

TOOL	FILE	VERSION
instmon	`instmon_v2_103_0_20261111.py`	2.103.0
Installer	`install_all_v7_5_0_20261110.sh`	7.5.0
Dashboard	`nodecontrol_pi02w_v9_3_112_20261111.py`	9.3.112-pi02w
SysMon	`sysmon_pi02w_v6_13_84_20261111.py`	6.13.84-pi02w
wifimon	`wifimon_v5_51_0_20261111.py`	5.51.0
44helper	`44helper_v0_1_19_20261108.py`	0.1.19
Node Talker (optional)	`nodetalker_v0_10_38_20261111.py`	0.10.38
Node Recorder (optional)	`noderecorder_v0_15_0_20261110.py`	0.15.0
Pibrary (optional)	`pibrary_v0_16_1_20261104.py`	0.16.1
Terminal Monitor (optional)	`termon_v0_2_4_20261109.py`	0.2.4
Uninstaller	`uninstall_all_v1_10_0_20261110.sh`	1.10.0

CONTENTS

In this manual

1 What gets installed: the tools, their ports and what the Pi02w build changes

2 Before you start: what the Pi needs, the root password, free memory

3 Step 1: get the files and install instmon

4 Step 2: install the rest with the installer

5 Step 3: open the pages and log in

6 Step 4: set up wifimon's WiFi list

7 Updating from GitHub with instmon

8 Memory on a 512 MB Pi: the launcher and its off switches

9 Troubleshooting

10 Removing the tools

11 Quick reference

12 Phone and its dialplan: what the Phone tab adds to Asterisk, and where the full guide is

How to read the commands

Commands are typed on the Pi, in an SSH session or on a keyboard and screen. Lines starting with `#` are notes, not commands. Replace `<pi-address>` with the Pi's IP address or host name.

```
# a note about the next line
sudo systemctl status sysmon
```

NOTE The scripts on GitHub are comment-stripped copies. Each tool's change history is in the `changelogs/` folder of the repo.

NOTE 44helper has its own manual, tab by tab: `docs/ASL-DVS_44helper_Manual.pdf`. The user manual covers every tool: `docs/ASL-DVS_User_Manual.pdf`.

What gets installed

Each tool is a single file. Each one runs as its own systemd service and serves its own page in the browser. instmon, the Dashboard, SysMon and wifimon are the core. 44helper, Node Talker, Node Recorder, Pibrary and Terminal Monitor are optional: a tool that is not installed is skipped unless you ask for it.

TOOL	SERVICE	PORT	WHAT IT DOES
Dashboard	nodecontrol	8989	Node control. Connect ASL and EchoLink nodes; tune DMR, YSF, P25, NXDN, D-STAR, XLX and M17; favorites; Phone.
SysMon	sysmon	9999	System monitor and health checks for the node, DVSwitch and its config files.
instmon	instmon	8990	Installer and manager. Install, update, start and stop the other tools; GitHub updates; Quiet System; disk images; the manuals.
wifimon	wifimon	8991	WiFi and supply-voltage watchdog. Powers the Pi off cleanly on sustained low voltage; reboots it (or powers it off) on a lost network.
44helper	44helper	9997	Optional. 44Net Connect, firewall, router and SvxLink helper.
Node Talker	nodetalker	8993	Optional. Listen to the node's audio in the browser and talk with push-to-talk; also the server for the Node Talker Mobile app.
Node Recorder	noderecorder	8995	Optional. Records the node's audio on a schedule to WAV, MP3, Opus or FLAC. HTTPS.
Pibrary	pibrary	9000	Optional. Media library for a USB or SD drive: movies, audio and PDFs in the browser.
Terminal Monitor	termon	7878	Optional, and never installed unless you ask. top, htop or btop and a root shell in the browser. HTTPS. Anyone on the LAN who knows the root password gets a root shell.

The Pi Zero 2 W build

SysMon and the Dashboard come in two builds. The Pi02w fork is the lighter one for the 512 MB Pi Zero 2 W. Its files have `_pi02w_` in the name and its `VERSION` ends in `-pi02w`.

DASHBOARD · PI02W

Keeps every tab, including Phone and all digital modes.

- Searches the AllStar and EchoLink lists from disk instead of memory.
- Polls Asterisk and refreshes the page less often.

SYSMON · PI02W

Drops the heaviest tabs:

- Console, Net, Reg, Firewall and Zello.
- Everything else matches the full build.

instmon, wifimon, 44helper and the installer are common: one build for every Pi.

NOTE The Watchdog script (`asl_dvs_watchdog`) is not needed. The Dashboard's own service already restarts it if it stops, and the installer removes an old Watchdog if it finds one.

Before you start

What the Pi needs

- A Raspberry Pi Zero 2 W already running AllStarLink with DVSwitch (Analog_Bridge and MMDVM_Bridge) set up.
- `python3` and `systemd`. Both come with Raspberry Pi OS.
- A network connection, and SSH access with `sudo`.

Set a root password

The web pages ask for a login, and they check it against the Pi's root password. Raspberry Pi OS ships with root locked, so set a password once:

```
sudo passwd root
```

Free up memory first

A Pi Zero 2 W has about 415 MB usable. Check what is free before you add tools:

```
free -m
```

Services you don't use are the easiest memory to win back. Skip any you do use:

SERVICE	TYPICAL USE	TURN OFF WITH
PackageKit	~12 MB swap	`sudo systemctl mask --now packagekit`
P25Gateway	~22 MB swap	`sudo systemctl disable --now p25gateway`
Allmon3	~44 MB swap	`sudo systemctl disable --now allmon3`

Figures from a working Pi Zero 2 W node. PackageKit only serves Cockpit's "Software updates" page; apt works without it. Undo with `sudo systemctl enable --now <name>` (or `unmask` for packagekit).

> **CAREFUL** Keep swap on. A Pi Zero 2 W needs it: turning swap off, or removing `dphys-swapfile`, runs the node out of memory. 44helper 0.0.195 removed its swap cards for that reason.

Get the files and install instmon

Get the files

Put all the files in one folder on the Pi, for example your home folder:

```
# files from the main branch on GitHub
cd ~
B=https://raw.githubusercontent.com/kd8pgk/ASL-DVS/main
wget $B/instmon_v2_103_0_20261111.py
wget $B/install_all_v7_5_0_20261110.sh
wget $B/nodecontrol_pi02w_v9_3_112_20261111.py
wget $B/sysmon_pi02w_v6_13_84_20261111.py
wget $B/wifimon_v5_51_0_20261111.py
# optional
wget $B/44helper_v0_1_19_20261108.py
wget $B/nodetalker_v0_10_38_20261111.py
wget $B/noderecorder_v0_15_0_20261110.py
wget $B/pibrary_v0_16_1_20261104.py
wget $B/termon_v0_2_4_20261109.py
```

TIP Newer files may be on GitHub by the time you read this. Check the repo's file list and use the newest `_pi02w_` SysMon and Dashboard files.

instmon goes first. Once it is running, it can install and update everything else from the browser, and it updates itself.

STEP 1 · INSTMON

```
sudo python3 instmon_v2_103_0_20261111.py --install
```

Then open `http://<pi-address>:8990` and log in as root.

What --install does

- Copies the file into place and writes `instmon.service`.
- Writes the shared launcher `/usr/local/bin/asl_dvs_launch.py` (see Memory, section 8).
- Enables the service so it starts on boot, and starts it now.

Optional: stage the other files in instmon

Instead of running the installer by hand (Step 2), you can upload the files on instmon's page, or let **Check GitHub** download them into instmon's library, and press **Install** there. The installer also finds files staged in instmon's library.

Check GitHub also downloads, but never installs, the newest file of each tool that is not installed. They show on the **Available, not installed** card: press **Install**, or tick several and press **Install selected**. Select all never ticks Terminal Monitor.

Install the rest with the installer

STEP 2 · INSTALLER

```
sudo bash install_all_v7_5_0_20261110.sh
```

What it asks

1. **Operation.** If tools are already installed: `i` install / reinstall, `u` update binaries only, or `q` quit. Config is kept either way.
2. **Version.** If more than one version of a tool is found, pick one (Enter takes the newest).
3. **Optional tools.** Node Talker, Node Recorder, Pibrary and Terminal Monitor are asked about only when they are not installed (the default is `n`). The Terminal Monitor question warns that it gives a root shell to anyone on the LAN who knows the password. A tool that is already installed is found, version-picked and updated like the others.
4. **First-run identity.** On a fresh node: your callsign, ASL node number and a display label. They go into `/etc/asl_dvs/asl_dvs.conf`.

Where it looks for files

The installer's own folder, the current folder, `/root`, `/home/pi`, your home folder, `/tmp` and instmon's library. Any subset of the tools can be installed: it installs what it finds. A tool with no file in those places is downloaded from `github.com/kd8pgk/ASL-DVS` into its folder in instmon's library. The download is the stable build by default (the file and commit listed in `channels/stable.json`, checked against its blob); a tool with no stable build yet is skipped with a warning. `ASL_DVS_CHANNEL=testing` takes every file from `main`, and `--offline` (or `--no-download`) turns downloads off. git is installed only if a download needs it.

If the download of an installed tool is older than the installed version, the installer says so and asks `Install the older version? [y/N]`. Answering no leaves the tool as it is.

Which build it picks

For SysMon and the Dashboard the build is chosen first, then only that build's files are offered:

1. `SYSMON_VARIANT` / `DASH_VARIANT` = `pi02w` or `full`, if set;
2. otherwise the build already installed;
3. otherwise `pi02w` on a Pi Zero 2 W, `full` on any other Pi.

```
# force the Pi02w build (sudo needs the variables on its own line)
sudo SYSMON_VARIANT=pi02w DASH_VARIANT=pi02w bash install_all_v7_5_0_20261110.sh
```

CAREFUL If no file of the chosen build is found, that tool is skipped with a warning. The other build is never used in its place.

Without questions

Add `--non-interactive` (or `-y`). It picks the newest files, updates an existing install, and takes the identity from environment variables on a fresh node:

```
sudo AUTO_CALLSIGN=KD8PGK AUTO_NODE=652701 AUTO_LABEL="Travel Node" \
     bash install_all_v7_5_0_20261110.sh --non-interactive
```

When it finishes, the installer checks that each service is running and listening on its port.

Optional tools and other flags

FLAG OR VARIABLE	WHAT IT DOES
INSTALL_NODETALKER=1	Installs Node Talker without asking. `INSTALL_NODERECORDER=1` and `INSTALL_PIBRARY=1` do the same for Node Recorder and Pibrary.
--with-termon	Installs Terminal Monitor (also `INSTALL_TERMON=1`, or a yes at the prompt). This is the only way a non-interactive run gets it.
--suite	Asks for Node Talker, Node Recorder and Pibrary without prompting. It never asks for Terminal Monitor.
--download-only	Fetches the files of every tool that is not installed into instmon's library for the chosen channel, and installs, enables and starts nothing. Install them later from the Available, not installed card.
--offline	No downloads: use only the files already on the Pi.
ASL_DVS_CHANNEL=testing	Take files from `main` instead of the stable list.

NOTE A service you stopped stays stopped. When a tool is already installed, its service is not running, and you stopped it in instmon (Stop) or it is disabled, the installer updates the file and then stops it again (and disables it again if it was disabled). The check at the end says "stopped, as you had it". A service that crashed, or one paused by Quiet System, is updated and started as before.

5 · STEP 3

Open the pages and log in

PAGE	ADDRESS
Dashboard	`http://<pi-address>:8989`
SysMon	`http://<pi-address>:9999`
instmon	`http://<pi-address>:8990`
wifimon	`http://<pi-address>:8991`
44helper	`http://<pi-address>:9997`
Node Talker (if installed)	`http://<pi-address>:8993`
Node Recorder (if installed)	`https://<pi-address>:8995`
Pibrary (if installed)	`http://<pi-address>:9000`
Terminal Monitor (if installed)	`https://<pi-address>:7878`

Log in as root with the root password from section 2. Node Recorder and Terminal Monitor use HTTPS with a self-signed certificate, so the browser warns once. The other pages use plain HTTP, so only open them on your own network or over a VPN.

> **TIP** On the first visit to the Dashboard, check that the callsign and node number in the header are right.

6 · STEP 4

Set up wifimon

On a first install, wifimon runs with both watchdogs off: it shows the WiFi and the voltage, but doesn't reconnect or shut the node down. When you log in to its page (port 8991) it asks "Turn the watchdogs on?". Add your WiFi networks first, then turn them on there or in the Watchdog card:

- **WiFi watchdog:** reconnects to saved networks. After 3 minutes with no connection it **reboots** the node (the default; at most 3 times in 6 hours, then it stays up and keeps trying) or powers it off, as you choose in the Watchdog card. It does not act in the first 10 minutes after boot, and it waits, with no timer, until at least one network is saved.
- **Voltage watchdog:** shuts the node down after 30 seconds of low voltage.

Updates and reinstalls keep the watchdogs as they were. Add the networks it may reconnect to:

```
sudo python3 /usr/local/bin/wifimon.py --setup-wifi
```

You can also manage networks and each WiFi adapter on wifimon's page (port 8991).

> **CAREFUL** wifimon powers the Pi off cleanly on sustained low voltage, and reboots it (or powers it off) on a sustained loss of network. Use a good power supply and cable: the Pi Zero 2 W is sensitive to voltage drop.

Updating from GitHub with instmon

The **GitHub Updates** card in instmon reads the repo's file list directly.

CHECK GITHUB

Shows the newest GitHub copy of each tool and whether it is newer than yours. Full and Pi02w builds get separate rows; rows for the build this Pi isn't using are tagged **other build**.

UPDATE

Downloads one file into instmon's library and checks it first: the same file GitHub listed, a version line, the right first line, a syntax check and `--install` support. Nothing installs until you press **Install**.

FULL UPDATE

1. Downloads and checks every new file before anything changes, and refreshes the downloads of tools that are not installed (it only downloads those).
2. Pauses the web tools.
3. One tool at a time: saves a copy of the installed file, uninstalls it, installs the new one. If the new one fails, it puts the old one back and moves on.
4. Updates instmon last. A timer reinstalls the old instmon if the new one isn't answering after 90 seconds.
5. Ends with a summary.

A tool you stopped (with Stop, or disabled) is updated and then stopped again. Nothing is changed while Node Recorder is recording: the buttons grey out and a popup says what is recording and until when.

ON A SLOW LINE

Check GitHub and the downloads wait longer (30, 60 then 90 seconds), skip a file that keeps stalling instead of ending the whole check, ask GitHub for gzip, and read only the top of each file to learn its version. The page itself is compressed, and polls less often when the line is slow (**Slow link: On** also drops the embedded fonts).

MANUALS

The Manuals card downloads only the manuals and guides (`ASL-DVS_*_Manual.pdf` and `ASL-DVS_*_Guide.pdf`), not the other PDFs in `docs/`.

The Pi Zero 2 W build choice

On a Pi Zero 2 W the GitHub Updates card shows a **Pi Zero 2 W build** choice: **Pi02w fork (default)** or **Full build (override)**. Full Update installs the chosen build, and switches an installed tool that doesn't match (listed as **SWITCH**).

Quiet System

Quiet System / Restore pauses every web tool except instmon and wifimon (SysMon, the Dashboard, 44helper, Node Talker, Node Recorder, Pibrary and Terminal Monitor) before a big upload or install. Asterisk, the bridges, Allmon3, SSH and wifimon keep running. Everything comes back by itself after 30 minutes.

Memory on a 512 MB Pi

The launcher

The SysMon and Dashboard, instmon, wifimon, 44helper, Node Talker, Node Recorder and Pibrary all start through `/usr/local/bin/asl_dvs_launch.py`:

```
ExecStart=/usr/bin/python3 /usr/local/bin/asl_dvs_launch.py /usr/local/bin/sysmon.py
```

Started directly, Python compiles the whole file on every start and keeps that memory. The launcher loads the tool as a module instead, so Python saves the compiled copy in `__pycache__` and reuses it. That is about half the memory, and twice as fast to start.

Three more memory savers

Launcher version 2 also turns these on. Each has an off switch:

SAVER	WHAT IT DOES	OFF SWITCH
-OO	Drops the built-in help text (docstrings) from the loaded code.	`/etc/asl_dvs/launch_no_optimize`
Pool cap	`MALLOC_ARENA_MAX=2`: at most 2 memory pools instead of up to 8 per CPU core.	`/etc/asl_dvs/launch_no_arena_cap`
Trim	1 minute after start, then every 5 minutes, hands freed memory back to Linux.	`/etc/asl_dvs/launch_no_trim`

To turn one off, create its file and restart the services. Delete the file and restart to turn it back on:

```
sudo touch /etc/asl_dvs/launch_no_arena_cap
sudo systemctl restart sysmon nodecontrol instmon wifimon
```

> **NOTE** The first start after an update compiles everything once more, so memory runs high. Measure after the next restart. With the first launcher version one node fell from about 390 to 294 of 415 MiB; the three savers haven't been measured on a Pi yet.

Checking memory

The Pi Zero 2 W uses zram: swap that is compressed in RAM. A program's real size is its RAM plus its swap:

```
for p in $(pgrep -f asl_dvs_launch.py); do
  printf "%-40s %s\n" "$(tr '\0' ' ' </proc/$p/cmdline | awk '{print $NF}')" \
    "$(grep -E 'VmRSS|VmSwap' /proc/$p/status | tr -s ' \n' ' ')"
done; free -m
```

Biggest swap users across the whole Pi:

```
for f in /proc/[0-9]*/status; do awk '/^Name:/{n=$2} /^VmSwap:/{if($2>0) printf "%7.1f MB
%s\n",$2/1024,n}' $f 2>/dev/null; done | sort -rn | head -10
```

Troubleshooting

A page doesn't load

```
# is the service running?
systemctl status sysmon --no-pager
# its last 30 log lines
journalctl -u sysmon -n 30 --no-pager
# is anything listening on the port?
ss -tlnp | grep 9999
```

Use the service name and port of the tool in question (table in section 1).

The login is refused

The pages check the root password. Set or reset it with `sudo passwd root`.

The installer says "No pi02w … build found"

The `_pi02w_` file for that tool isn't in any of the folders it searches. Put it next to the installer and run it again.

The Pi is slow or runs out of memory

- Run the memory commands in section 8 and look for the biggest swap users.
- Check that every tool line shows `asl_dvs_launch.py`. A tool started as plain `python3 tool.py` uses about twice the memory: run its `--install` again.
- Check that swap is on: `swapon --show` should list it. If an older 44helper turned it off, use 44helper's System Optimization tab (**Swap -- restore**), or `sudo systemctl enable --now dphys-swapfile`.
- Close Cockpit's page when you're done: its two `cockpit-bridge` processes use about 27 MB while it is open.

A tool misbehaves after an update

Turn the memory savers off one at a time (section 8), starting with the pool cap, and restart. If that doesn't help, reinstall the previous version from instmon's library.

SysMon's own checks

```
sudo python3 /usr/local/bin/sysmon.py --selftest
```

It runs SysMon's security checks against this Pi and prints PASS / WARN / FAIL for each, without starting the web page.

Removing the tools

Remove one tool with its own uninstall:

```
sudo python3 /usr/local/bin/sysmon.py --uninstall
```

Remove every tool except instmon:

```
sudo bash uninstall_all_v1_10_0_20261110.sh
```

The script also removes Terminal Monitor when it is installed (its settings, certificate and `termon` user stay). It does not remove Node Recorder or Pibrary: use their Uninstall button in instmon, or their own `--uninstall`.

This leaves `/etc/asl_dvs/`, instmon's library and `/etc/wifimon/wifimon.conf` in place. Remove instmon last with its own `--uninstall`. The launcher is removed once no service uses it.

Quick reference

TASK	COMMAND
Install instmon	`sudo python3 instmon_v2_103_0_20261111.py --install`
Install the rest	`sudo bash install_all_v7_5_0_20261110.sh`
Set the login password	`sudo passwd root`
Add WiFi networks	`sudo python3 /usr/local/bin/wifimon.py --setup-wifi`
Restart the web tools	`sudo systemctl restart sysmon nodecontrol instmon wifimon`
A tool's log	`journalctl -u <service> -n 30 --no-pager`
Memory overview	`free -m · zramctl`
Saver off switch	`sudo touch /etc/asl_dvs/launch_no_<optimize|arena_cap|trim>`
Remove all but instmon	`sudo bash uninstall_all_v1_10_0_20261110.sh`

SERVICE	PORT	CONFIG
nodecontrol	8989	/etc/asl_dvs/asl_dvs.conf
sysmon	9999	/etc/sysmon/sysmon.conf
instmon	8990	/etc/asl_dvs/ · library in /etc/asl_dvs/instmon_library
wifimon	8991	/etc/wifimon/wifimon.conf
44helper	9997	/etc/44helper/44helper.conf
nodetalker, noderecorder	8993, 8995	/etc/asl_dvs/nodetalker.conf, /etc/asl_dvs/noderecorder/
pibrary, termon	9000, 7878	/etc/pibrary/pibrary.conf, /etc/asl_dvs/termon.conf

Phone and its dialplan

The Dashboard's Phone tab places and answers calls through Asterisk, the phone engine under AllStarLink. Nothing changes in Asterisk until you tick **Activate** on the Edit page (Phone section). After that, every **Save Phone** rewrites the Dashboard's own files and reloads Asterisk.

What Activate adds

FILE IN /ETC/ASTERISK	WHAT THE DASHBOARD ADDS
`rpt.conf`	One private node per phone network (1000–1999), between its `dvs-phone` marker lines.
`extensions.conf`	One line at the end: `#tryinclude dvs_phone_extensions.conf` (the phone dialplan).
`pjsip.conf / iax.conf`	Include lines for the SIP and IAX2 account files.
`modules.conf`	Load lines for the phone modules. New SIP modules need one Asterisk restart.
`/var/lib/asterisk`	Three small scripts: hang up, radio tone codes, radio dial requests.

From the radio

CODE	DOES
`*61 + number`	Dial. The Dashboard places the call (the Phone tab must be open).
`*62 or *65`	Hang up.
`*63 / *64`	Phone patch on / off.
`*980 *981 *982`	Send `*`, `#` or `*99` into the call.

How calls reach the radio

Each phone network has its own node, linked to your radio node only while the Phone tab is open. Calls join that node with voice keying (`rpt(node,Pv)`), so a simplex radio transmits only while the far end talks and can hear you in the gaps. Tick **Listen only** to link the phone node one-way: you hear calls, and nothing from the radio goes into them.

```
# is the phone node linked, and which way? (R = listen only, T = two-way)
sudo asterisk -rx "rpt xnode <your-node>" | grep ALINKS
# the dialing rules Asterisk has loaded for one network
sudo asterisk -rx "dialplan show dvs-net-<network-id>"
```

TIP SysMon's Phone tab reads all of this from the files, read-only, and checks it: nodes, networks, sign-in, dialing rules and the modules the phone needs.

NOTE The nuts and bolts (dialplan syntax, patterns, every context the Dashboard writes, autopatch, call flows and troubleshooting) are in `docs/ASL-DVS_Pi02w_Phone_Dialplan_Guide.pdf`.